Om Raj Singh

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Summary

M.Tech student specializing in Data Science and Artificial Intelligence with hands-on experience in deep learning, computer vision, adversarial machine learning, and LLM-based tool-calling agents. Skilled in building PyTorch-based machine learning pipelines, implementing research-oriented AI systems, evaluating models using task-specific metrics, and writing clean, modular Python code.

Education

Indian Institute of Technology Bhilai **[2025 – 2027]**
M.Tech in Data Science and Artificial Intelligence CGPA: [7.3]

Lakshmi Narain College of Technology Bhopal **[2021 – 2025]**
B.Tech in CSE - AIML CGPA: [7.8]

Skills

Programming Languages: Python, C++

Machine Learning and Deep Learning: PyTorch, scikit-learn, model evaluation, feature engineering, supervised learning, unsupervised learning

LLM and Generative AI: Transformers, Hugging Face, RAG, embeddings, vector databases, prompt engineering, tool calling, LLM agents

Computer Vision: OpenCV, CNNs, Vision Transformers

Data and Scientific Computing: NumPy, Pandas, Matplotlib, Seaborn

Tools and Platforms: Git, GitHub, Linux, VS Code, Jupyter Notebook

Projects

MiniGaker: Adversarial Image Generation and Evaluation Framework *Python, PyTorch*

- Built a PyTorch-based framework to generate and evaluate adversarial examples against ImageNet-trained classification models.
- Tested targeted and untargeted attacks on victim models including ResNet50, ResNet152, and DenseNet121.
- Implemented evaluation metrics such as targeted ASR top-1, targeted ASR top-5, untargeted ASR, clean accuracy, and adversarial accuracy.

LLM Tool-Calling Agent *Python, Google GenAI SDK*

- Built a multi-turn LLM agent with external tool-calling support using the Google GenAI SDK.
- Designed custom Python tools with structured descriptions, input arguments, return formats, and error-handling logic.
- Integrated external APIs using Python `requests`, environment variables, and JSON response parsing.

Retina-SEM: Unsupervised Retinal Vessel Segmentation *Python, OpenCV, NumPy, scikit-learn, Matplotlib*

- Implemented an unsupervised retinal vessel segmentation pipeline using structural entropy minimization over image graph representations.
- Built graph-based and multi-scale image representations to identify vessel structures without pixel-level manual labels.
- Evaluated segmentation quality using Dice score, clDice, precision, recall, and PR-AUC.

Achievements

- GATE CS 2024 Rank: 3754
- CAT 2024: 86.21 %ile
- GATE DSAI 2025 Rank 6991